

Supplementary Experimental Configuration for RQ1, RQ2, and RQ3

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1 RQ1 TeaStore

Table 1. Five Scenarios' Configurations of TeaStore by Differentiating Configurations at T_1

Scenario	Microservice implementation	Availability	Response time	Estimates at T_1			
				Vulnerabilities			
s_1	Webui ₀	0.9	100	CVE-2023-34055	CVE-2024-38816	CVE-2025-24813	
	Authentication ₀	0.92	400	CVE-2023-34055	CVE-2025-22228	CVE-2023-46589	
	Persistence ₀	0.85	300	CVE-2023-34055	CVE-2024-34447	CVE-2024-22262	CVE-2023-2976
	Image ₀	0.92	400	CVE-2023-34055	CVE-2024-38809	CVE-2024-22262	
	Recommender ₀	0.9	450	CVE-2023-34055	CVE-2024-38286	CVE-2024-34447	
s_2	WebUI ₀	0.9	100	CVE-2023-34055	CVE-2024-38816	CVE-2025-24813	CVE-2024-34447
	Authentication ₀	0.92	400	CVE-2023-34055	CVE-2025-22228	CVE-2023-46589	
	Persistence ₀	0.85	300	CVE-2023-34055	CVE-2024-34447	CVE-2024-22262	CVE-2023-2976
	Image ₀	0.92	400	CVE-2023-34055	CVE-2024-38809	CVE-2024-22262	
	Recommender ₀	0.9	450	CVE-2023-34055	CVE-2024-38286	CVE-2024-34447	
s_3	WebUI ₀	0.85	200	CVE-2023-34055	CVE-2024-38816	CVE-2024-34447	
	Authentication ₀	0.85	400	CVE-2023-34055	CVE-2025-22228	CVE-2023-46589	CVE-2023-2976
	Persistence ₀	0.75	400	CVE-2023-34055	CVE-2024-34447	CVE-2024-22262	
	Image ₀	0.75	350	CVE-2023-34055	CVE-2024-38809	CVE-2024-22262	
	Recommender ₀	0.9	450	CVE-2023-34055	CVE-2024-38286	CVE-2024-34447	
s_4	WebUI ₀	0.9	50	CVE-2023-34055	CVE-2024-38816	CVE-2024-34447	
	Authentication ₀	0.8	200	CVE-2023-34055	CVE-2025-22228	CVE-2023-46589	
	Persistence ₀	0.7	300	CVE-2023-34055	CVE-2024-34447	CVE-2024-22262	CVE-2023-2976
	Image ₀	0.6	400	CVE-2023-34055	CVE-2024-38816	CVE-2024-34447	
	Recommender ₀	0.9	450	CVE-2023-34055	CVE-2024-38286	CVE-2024-34447	
s_5	WebUI ₀	0.95	50	CVE-2023-34055	CVE-2024-38816	CVE-2024-34447	
	Authentication ₀	0.92	100	CVE-2023-34055	CVE-2024-12798	CVE-2024-38816	
	Persistence ₀	0.85	300	CVE-2023-34055	CVE-2024-34447	CVE-2023-46589	
	Image ₀	0.9	200	CVE-2023-34055	CVE-2024-47072	CVE-2025-22228	CVE-2024-22262
	Recommender ₀	0.9	450	CVE-2023-34055	CVE-2024-38286	CVE-2024-34447	

Table 2. Available Implementation Options at T_1 for Different Scenarios of TeaStore

Scenario / Microservice	Implementation option	Availability	Response time	Estimates at T_1			
				Vulnerabilities			
s_1 / WebUI	WebUI ₁	0.92	80	CVE-2023-34055	CVE-2024-38816		
	WebUI ₂	0.96	60	CVE-2023-34055	CVE-2024-38816	CVE-2023-44487	CVE-2025-24813
s_2 / WebUI	WebUI ₁	0.96	10	CVE-2023-34055	CVE-2025-24813	CVE-2024-34447	
	WebUI ₂	0.95	30	CVE-2023-34055	CVE-2024-34447		
s_3 / Authentication	Authentication ₁	0.9	400	CVE-2023-34055	CVE-2025-22228	CVE-2023-46589	CVE-2023-2976
	Authentication ₂	0.85	300	CVE-2023-34055	CVE-2025-22228	CVE-2023-46589	CVE-2023-2976
	Authentication ₃	0.85	400	CVE-2023-34055	CVE-2025-22228	CVE-2023-33202	
s_4 / Persistence	Persistence ₁	0.72	300	CVE-2023-34055	CVE-2024-34447	CVE-2024-22262	CVE-2023-2976
	Persistence ₂	0.7	260	CVE-2023-34055	CVE-2024-34447	CVE-2024-22262	CVE-2023-2976
	Persistence ₃	0.7	300	CVE-2023-34055	CVE-2024-34447		CVE-2025-24813
s_5 / Image	Image ₁	0.98	400	CVE-2023-34055	CVE-2024-47072	CVE-2025-22228	CVE-2024-22262
	Image ₂	0.6	150	CVE-2023-34055	CVE-2024-47072	CVE-2025-22228	CVE-2024-22262
	Image ₃	0.7	300	CVE-2023-34055	CVE-2024-47072	CVE-2024-38816	CVE-2025-24813

2 RQ2 Scenarios for SEFA and TeaStore

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Table 3. Three Scenarios' Configurations of SEFA by Differentiating Configurations at T_1

Scenario	Microservice implementation	Estimates						
		Availability	Response time	Vulnerabilities at T_1				
s_1	Restaurant ₀	0.75	200	CVE-2022-1471	CVE-2025-22228	CVE-2024-34447	CVE-2024-38286	CVE-2025-24813
	Ordering ₀	0.9	150	CVE-2024-12798	CVE-2024-38809	CVE-2024-38820		
	Delivery proxy ₀	0.85	100	CVE-2024-12798	CVE-2024-38809	CVE-2023-33202		
	Payment proxy ₀	0.95	50	CVE-2024-12798	CVE-2024-38286	CVE-2023-33202		
s_2	Restaurant ₀	0.9	50	CVE-2024-12798	CVE-2024-38820			
	Ordering ₀	0.85	150	CVE-2024-12798	CVE-2024-38820	CVE-2022-1471		
	Delivery proxy ₀	0.95	100	CVE-2024-12798	CVE-2024-38820			
	Payment proxy ₀	0.8	300	CVE-2024-38820	CVE-2024-34447	CVE-2022-1471	CVE-2025-22228	CVE-2023-6378
s_3	Restaurant ₀	0.95	100	CVE-2024-38286	CVE-2024-38820	CVE-2024-12798		
	Ordering ₀	0.85	100	CVE-2024-38286	CVE-2024-38820	CVE-2022-1471	CVE-2025-24813	CVE-2025-22228
	Delivery proxy ₀	0.9	50	CVE-2024-38286	CVE-2024-38820	CVE-2022-40150		
	Payment proxy ₀	0.9	150	CVE-2024-38286	CVE-2024-38820	CVE-2023-6378		

Table 4. Three Scenarios' Configurations of TeaStore by Differentiating Configurations at T_1

Scenario	Microservice implementation	Estimates						
		Availability	Response time	Vulnerabilities at T_1				
s_1	WebUI ₀	0.8	180	CVE-2024-47072	CVE-2024-38816	CVE-2025-24813	CVE-2024-34447	
	Authentication ₀	0.95	130	CVE-2024-38820	CVE-2022-1471	CVE-2024-12798		
	Persistence ₀	0.9	50	CVE-2024-38809	CVE-2024-12798	CVE-2024-38820		
	Image ₀	0.85	150	CVE-2022-1471	CVE-2024-38820	CVE-2024-12798		
	Recommender ₀	0.8	200	CVE-2022-1471	CVE-2024-12798	CVE-2024-22243		
s_2	WebUI ₀	0.95	50	CVE-2025-22228	CVE-2024-34447	CVE-2024-38820		
	Authentication ₀	0.9	100	CVE-2024-12798	CVE-2024-22243	CVE-2025-22228	CVE-2023-46589	CVE-2023-34055
	Persistence ₀	0.9	150	CVE-2025-22228	CVE-2023-46589			
	Image ₀	0.8	200	CVE-2023-46589	CVE-2024-38816			
	Recommender ₀	0.8	150	CVE-2024-34447	CVE-2024-38820	CVE-2024-38816		
s_3	WebUI ₀	0.9	50	CVE-2024-22262	CVE-2024-38809			
	Authentication ₀	0.95	30	CVE-2024-22259	CVE-2023-33202			
	Persistence ₀	0.95	80	CVE-2025-22228	CVE-2024-38809	CVE-2024-47072	CVE-2022-1471	CVE-2023-33202
	Image ₀	0.85	100	CVE-2024-38809	CVE-2024-22259	CVE-2024-38820		
	Recommender ₀	0.9	80	CVE-2024-22262	CVE-2024-38820	CVE-2025-22228		